

Network Programming

BCA — Semester VI — 2023 — Answer Sheet

Subject Code: CACS355

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Define the client-server model of a network application. Explain the design considerations for a network application.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Write a program to show the IP and MAC address of the system and also check if the IP address is IPv4 or IPv6.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Define URL Connection Class? Write a program to read header using header-specific methods.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write the functions of Socket. Explain the configuration options of a TCP client socket.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define a secure socket. Write a program to create a secure client socket.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define Buffer. Explain buffer handling methods.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes on (any two): a) HTTP methods b) Datagram Packet c) Multicast Socket

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions [2x10=20]

9. a) Differentiate between URL and URI class with their applications. b) Write any program to illustrate the application of RMI.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Write a TCP multithreaded server and client socket program to check the number sent by the client is either prime or composite.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Write a program to create a UDP server and client which converts the date from the client in BS to AD format. [Assume that the difference between BS and AD is 56 years, 08 months, 17 days]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.